AWS Day 2 Operations & Troubleshooting · NorthBank

SEV-1

LAB 01: The First 5 Minutes

Mastering the baseline reflexes of incident response.

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SEV-1

Reporter: Marco (App team)

Customers can’t reach internet banking — site returns 503.

```
$ export ALB_DNS, ASG_NAME (from
terraform output)

$ ./break.sh
```

Behind the Scenes: The `./break.sh` script silently sets desired capacity to 0, killing both servers. The objective is to diagnose and resolve the 503 error within 5 minutes without knowing this underlying script action.

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The Golden Rule of Day 2 Operations

The Amateur

Mindset

What's broken?

Action

Guesses randomly based on symptoms.

Execution

Changes configurations to see what happens.

Result

Wastes time checking perfectly healthy systems.

The Pro

Mindset

What changed?

Action

Reads the architecture front-door-in.

Execution

Queries CloudTrail (The Money Question).

Result

Formulates and tests exactly one hypothesis at a time.

What changed? always beats **What's broken?**

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The 5 Rules of Engagement



1. Confirm Identity

Who/where am I? (Never touch anything until identity is verified).

2. Find the Delta

What changed? (`CloudTrail` is the money question).

3. Follow the Path

Read the symptom front-door-in. (No random guessing).

4. State a Hypothesis

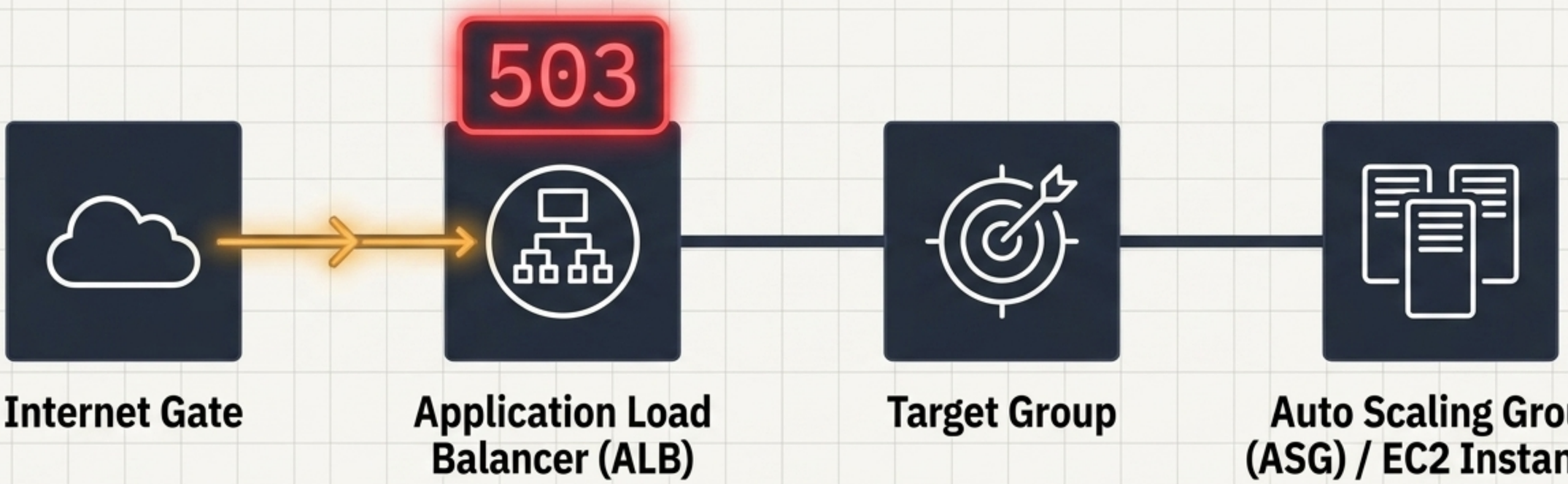
One theory, out loud, then test it.

5. Surgical Action

Smallest fix, then verify and communicate.

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“Front-Door-In” Traffic Flow Investigation



To fix the 503, we don’t jump straight to the servers. We start at the literal **front door** (ALB) and walk the architecture backward until we find the failure point.

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Rules Applied: Confirm Identity & Verify Symptom

Action:

We must replicate the user's issue to confirm the ticket, and then verify our **AWS execution role** to ensure we have the right access context.

Always confirm identity first.

```
> curl $ALB_DNS

[Output] 503 Service Temporarily Unavailable

> aws sts get-caller-identity
[Output] Arn: arn:aws:iam::123456789012:user/SRE-Admin
```

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Rules Applied: Follow the Path & State a Hypothesis

Hypothesis:

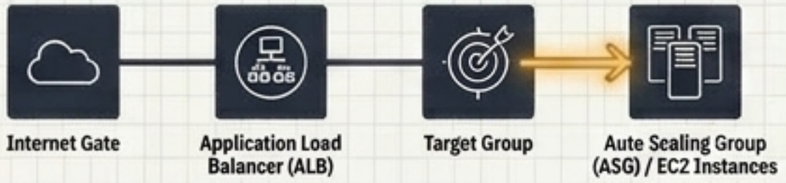
If the ALB returns a 503, the targets behind it might be dead or missing. We need to check the Target Group health.

Result: No healthy targets registered. The failure point is confirmed at the instance layer.

```
> aws elbv2 describe-target-health
--target-group-arn $TG_ARN

[Output] { TargetHealthDescriptions: [] }
```

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Rule Applied: Find the Delta (The Money Question)

Hypothesis:

Targets don’t just disappear. Something changed the ASG state. We check CloudTrail for recent API calls.

The Smoking Gun found via CloudTrail.

```
> aws cloudtrail lookup-events
--lookup-attributes
AttributeKey=EventName,AttributeValue=UpdateAuto
ScalingGroup

[Output] EventName: UpdateAutoScalingGroup,
RequestParameters: {desiredCapacity: 0, minSize:
0}
```

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Rule Applied: Surgical Action

Strategy:

Action: Reverse the exact unauthorized change discovered in CloudTrail, wait for target registration, and verify.

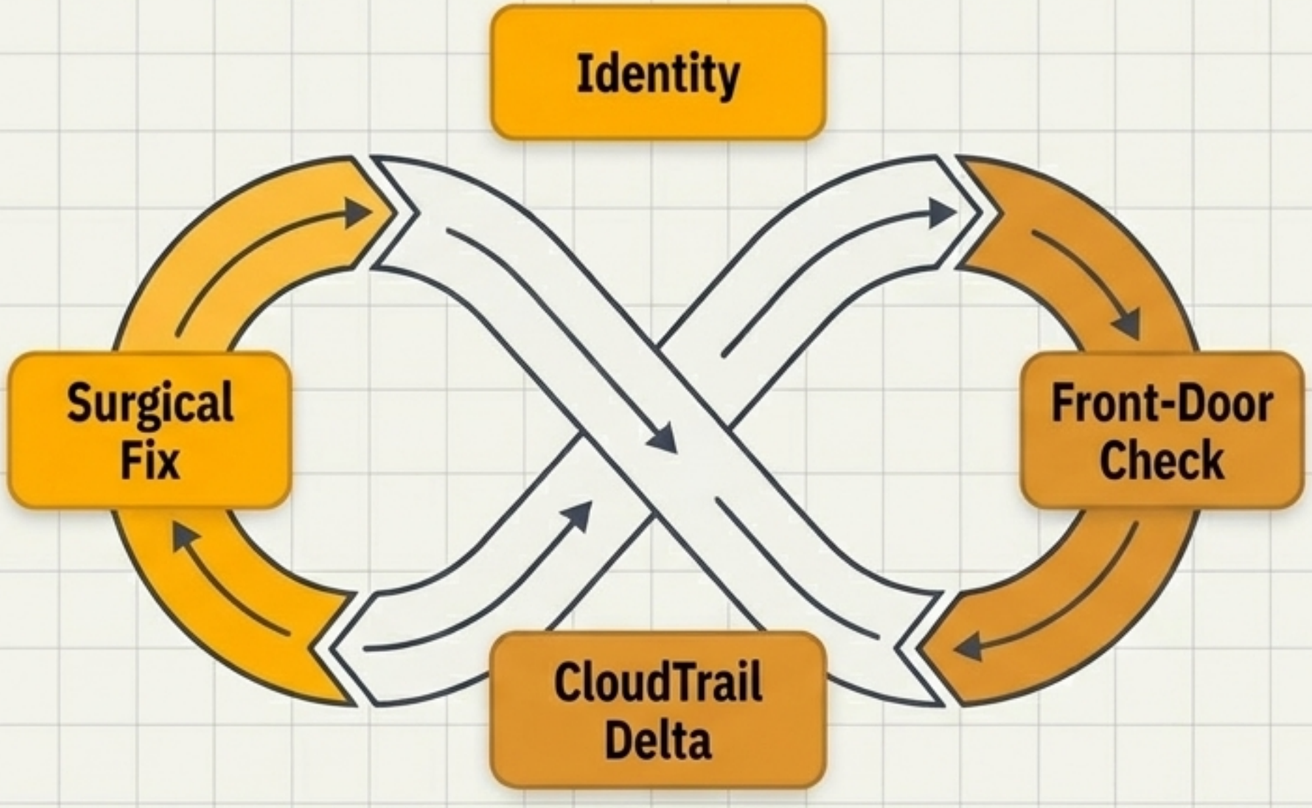
Resolution achieved by precision, not guesswork.

```
> aws autoscaling update-auto-scaling-group
> aws autoscaling update-auto-scaling-group
--auto-scaling-group-name $ASG_NAME
--min-size 2 --desired-capacity 2

> curl $ALB_DNS   (Wait ~30 seconds)
[Output] 200 OK
```

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The Blueprint for Everything That Follows



Lab 01 is not just about fixing a 503 error. It establishes the investigation **spine**.

This exact methodology repeats in **every NorthBank** lab from here on.

Always check caller identity first.

Find what changed via CloudTrail; apply the smallest possible fix.

The Day-2 Lesson:
What changed? beats what's broken?

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